Unsupervised Learning: Clustering

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Based on the slides of Nick Rouzzi and Vibhav Gogate

Clustering

Clustering systems:

- Unsupervised learning
- Requires data, but no labels
- Detect patterns, e.g., in
  - Group emails or search results
  - Customer shopping patterns
- Useful when don't know what you're looking for...
  - But often get gibberish

Clustering

- Want to group together parts of a dataset that are close together in some metric
  - Useful for finding the important parameters/features of a dataset

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Clustering

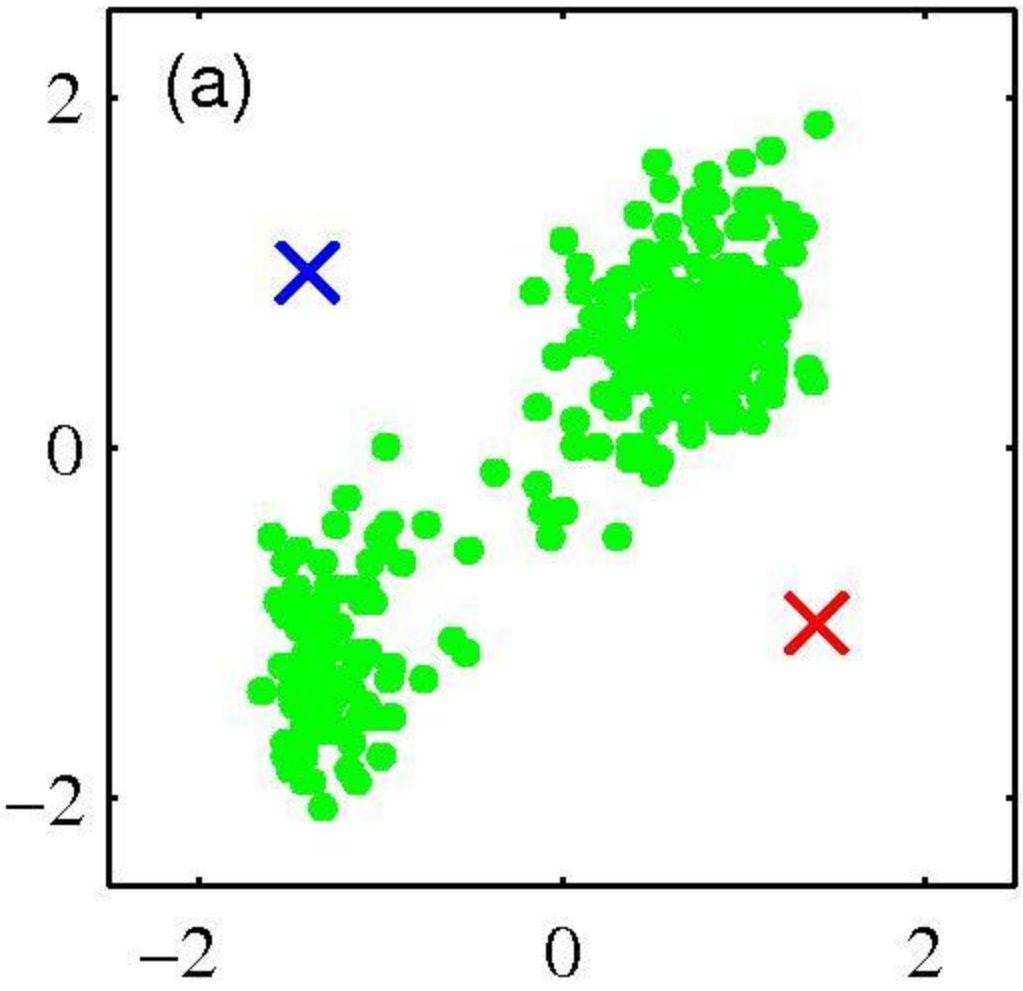
- Input: a collection of points $x^{(1)}, \ldots, x^{(m)} \in \mathbb{R}^n$, an integer k

- Output: A partitioning of the input points into k sets that minimizes some metric of closeness

k-means Clustering

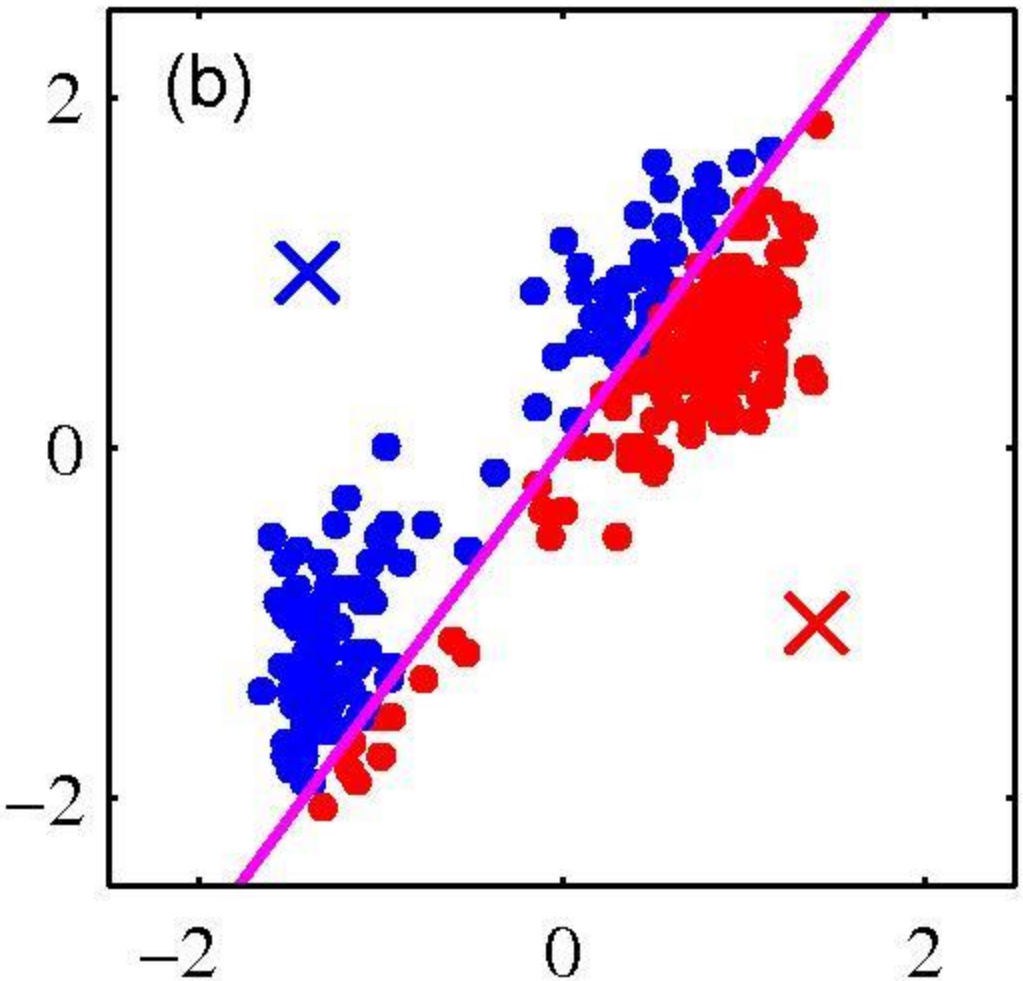
- Pick an initial set of k means (usually at random)
- Repeat until the clusters do not change:
  - Partition the data points, assigning each data point to a cluster based on the mean that is closest to it
  - Update the cluster means so that the ith mean is equal to the average of all data points assigned to cluster i

k-means clustering: Example



Pick k random points as cluster centers (means)

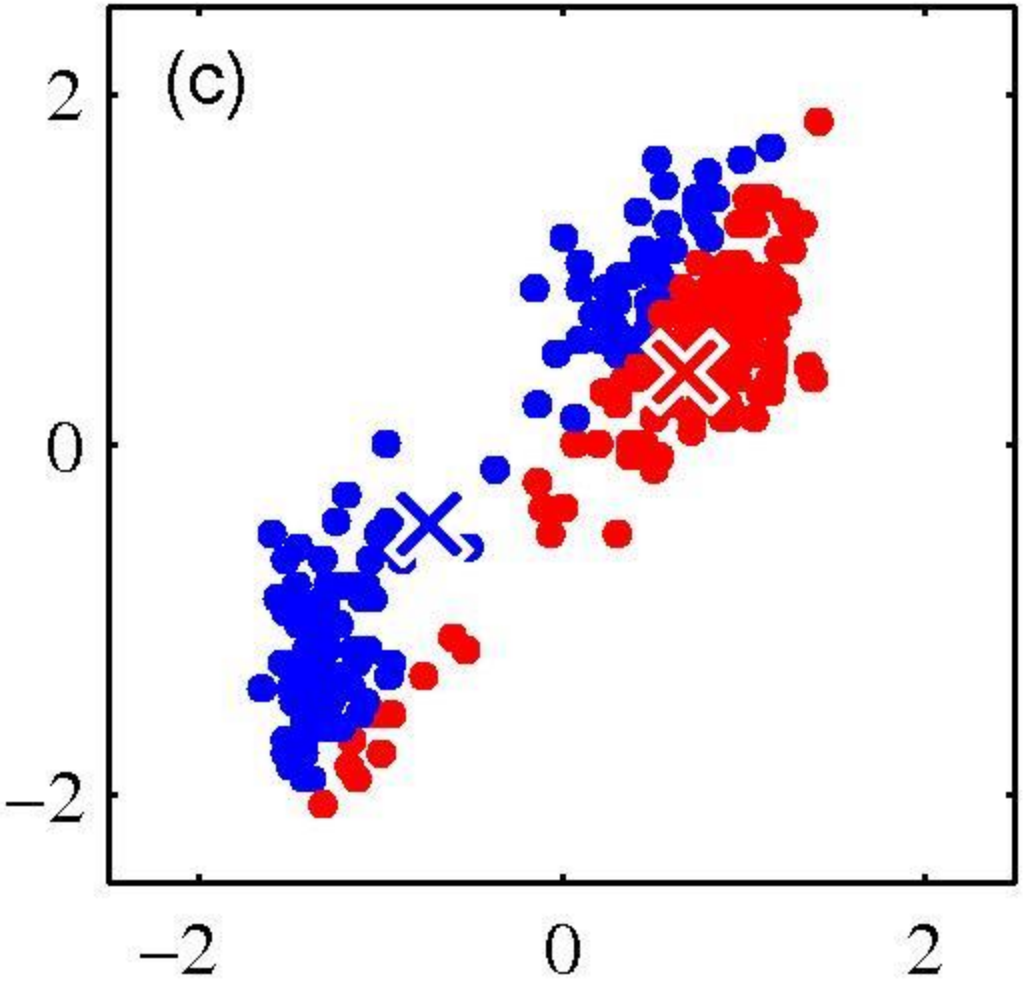
k-means clustering: Example



Iterative Step 1:

Assign data instances to closest cluster center

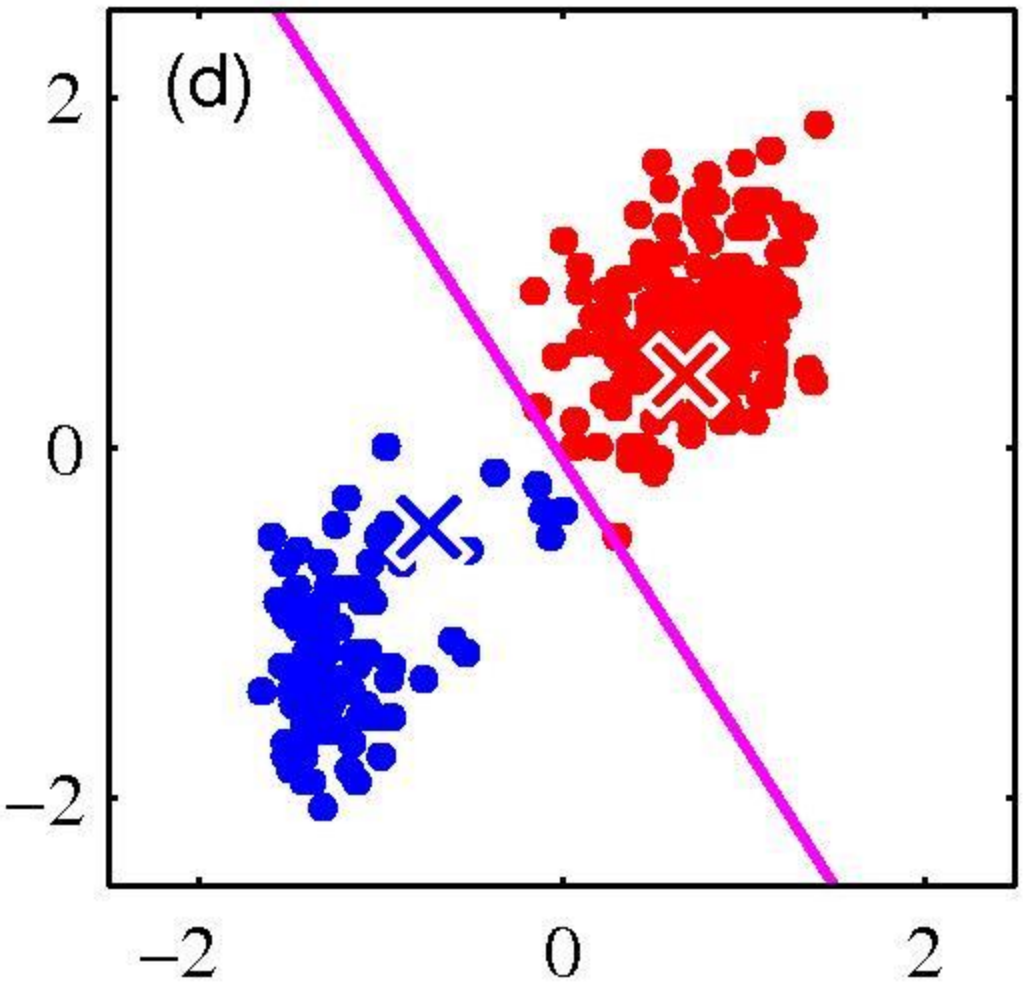
k-means clustering: Example



Iterative Step 2:

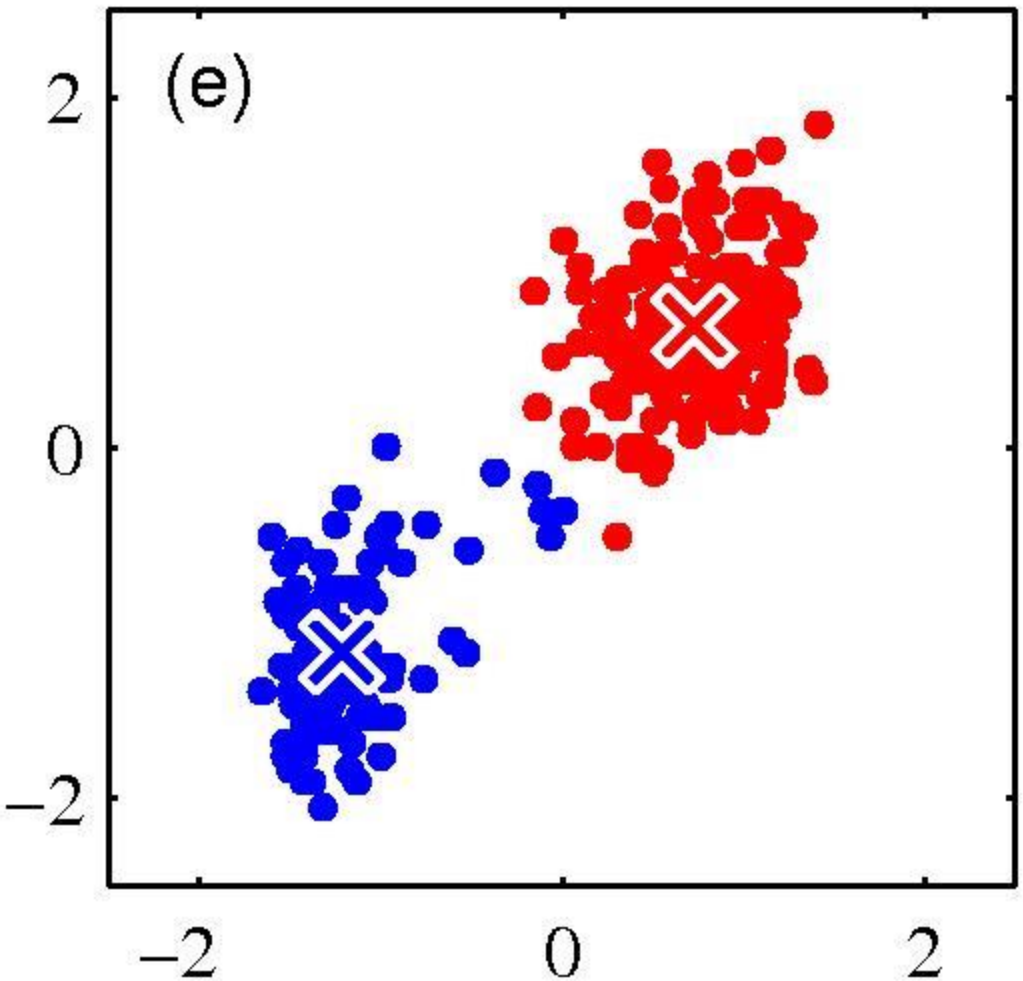
Change the cluster center to the average of the assigned points

k-means clustering: Example

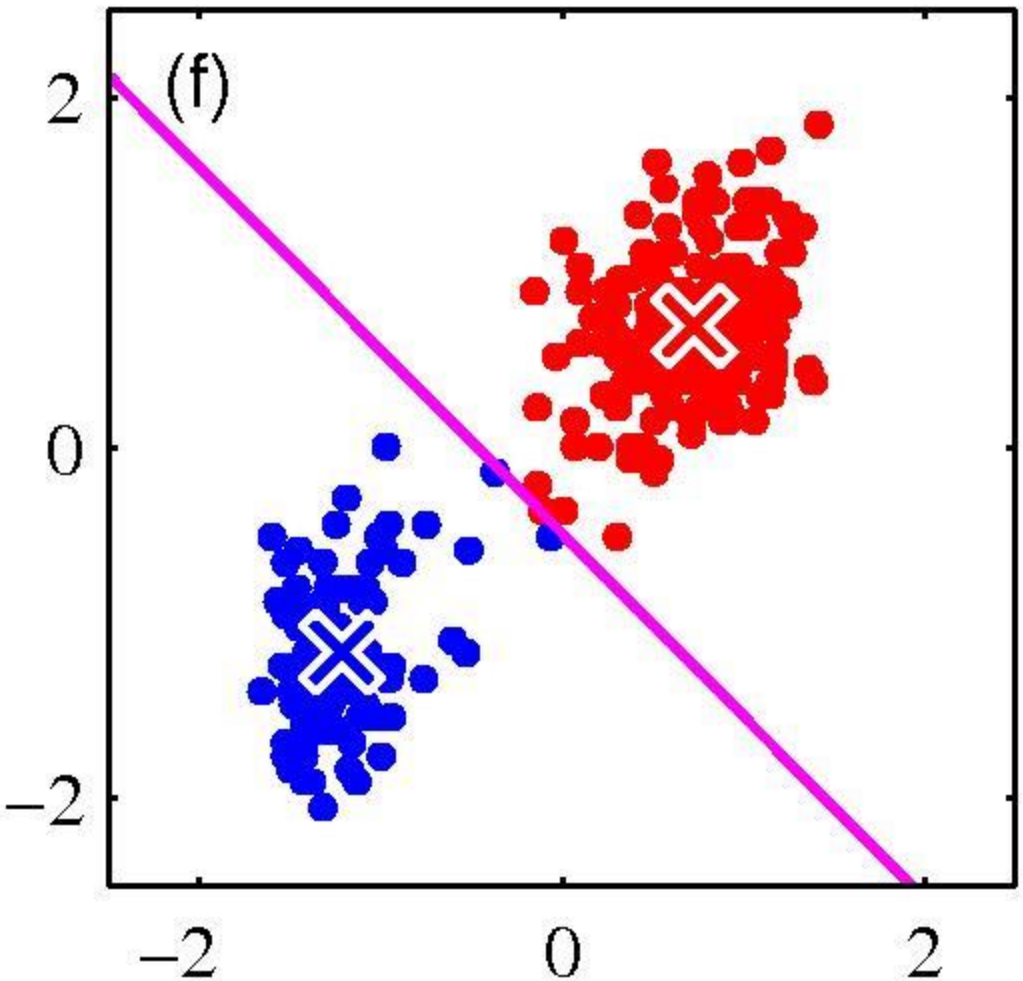


Repeat until convergence

k-means clustering: Example



k-means clustering: Example



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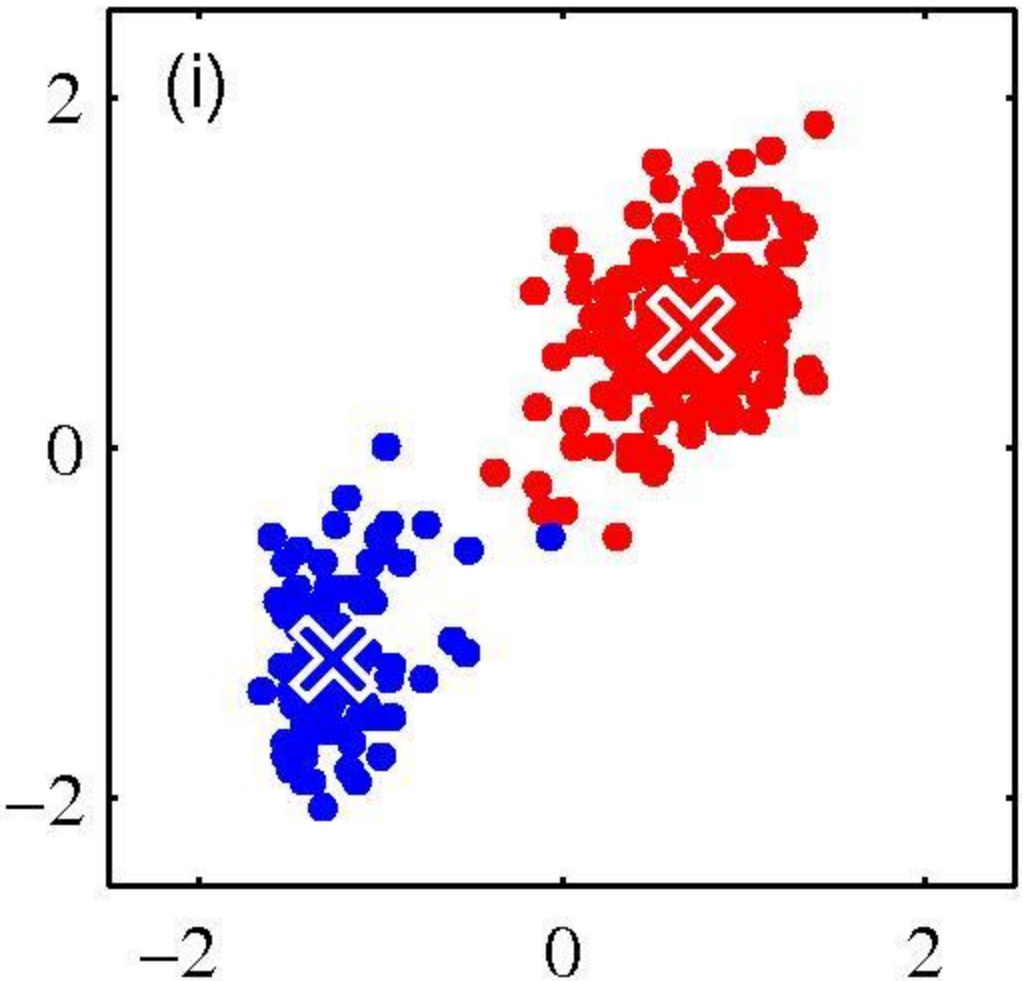

k-Means for Segmentation

$k = 2$

Goal of segmentation is to partition an image into regions, each of which has reasonably homogenous visual appearance

Original

k-Means for Segmentation

$k = 2$

$k = 3$

Original

k-Means for Segmentation

$k = 2$

$k = 3$

$k = 10$

Original

k-means Clustering as Optimization

- Minimize the distance of each input point to the mean of the cluster/partition that contains it

$$\min_{S_1,\ldots,S_k} \sum_{i=1}^{k} \sum_{j \in S_i} \left\| x^{(j)} - \mu_i \right\|^2$$

where

- $S_i \subseteq \{1, \ldots, M\}$ is the ith cluster
- $S_i \cap S_j = \emptyset$ for $i \neq j$, $\cup_i S_i = \{1, \ldots, n\}$
- μ_i is the centroid of the ith cluster

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Exactly minimizing this function is NP-hard (even for $k = 2$)

k-means Clustering

- The k-means clustering algorithm performs a block coordinate descent on the objective function

$$\sum_{i=1}^{k} \sum_{j \in S_i} \left\| x^{(j)} - \mu_i \right\|^2$$

- This is not a convex function: could get stuck in local minima

k-Means as Optimization

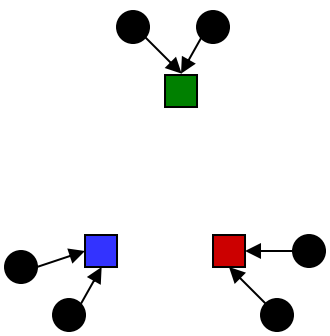
- Consider the k-means objective function

$$\phi(x, S, \mu) = \sum_{i=1}^{k} \sum_{j \in S_i} \left\| x^{(j)} - \mu_i \right\|^2$$

points

cluster assignments

cluster means



- Two stages each iteration
  - Update cluster assignments: fix means μ, change assignments S
  - Update means: fix assignments S, change means μ

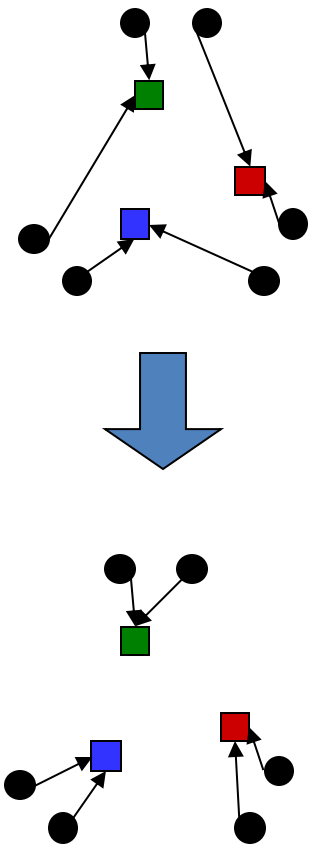
Phase I: Update Assignments

- For each point, re-assign to closest mean, $x^{(j)} \in S_i$ if

$$j \in \arg\min_i \left\| x^{(j)} - \mu_i \right\|^2$$

- Can only decrease ϕ as the sum of the distances of all points to their respective means must decrease

$$\phi(x, S, \mu) = \sum_{i=1}^{k} \sum_{j \in S_i} \left\| x^{(j)} - \mu_i \right\|^2$$

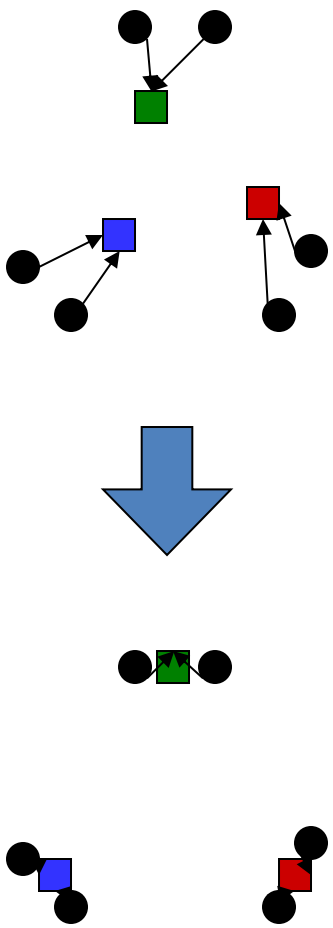


Phase II: Update Means

- Move each mean to the average of its assigned points

$$\mu_i = \sum_{j \in S_i} \frac{x^{(j)}}{|S_i|}$$

- Also can only decrease total distance…
  - Why?



Phase II: Update Means

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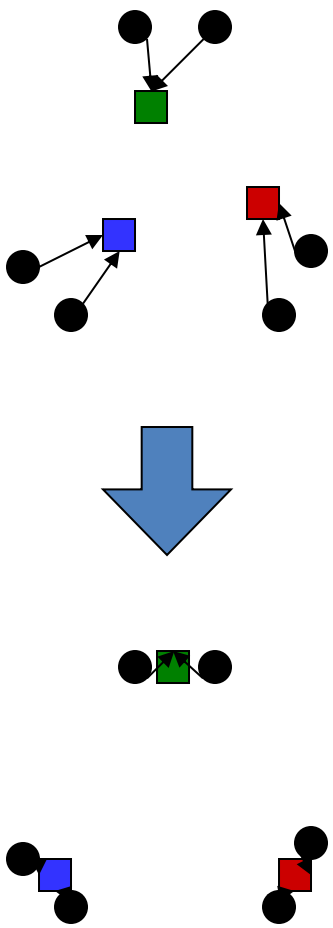
- Also can only decrease total distance…
  - The point y with minimum squared Euclidean distance to a set of points is their mean



Initialization

- K-means is sensitive to initialization
  - It does matter what you pick!
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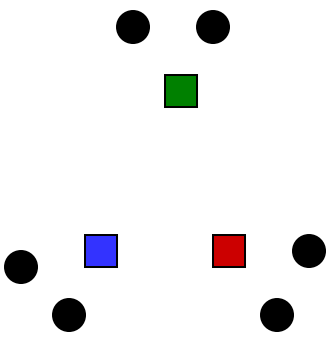
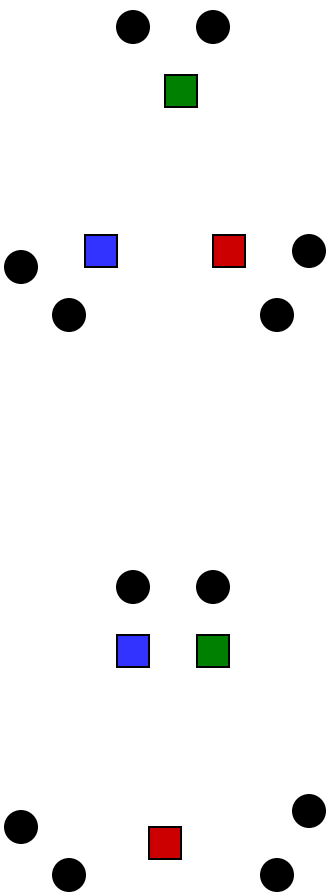


Initialization

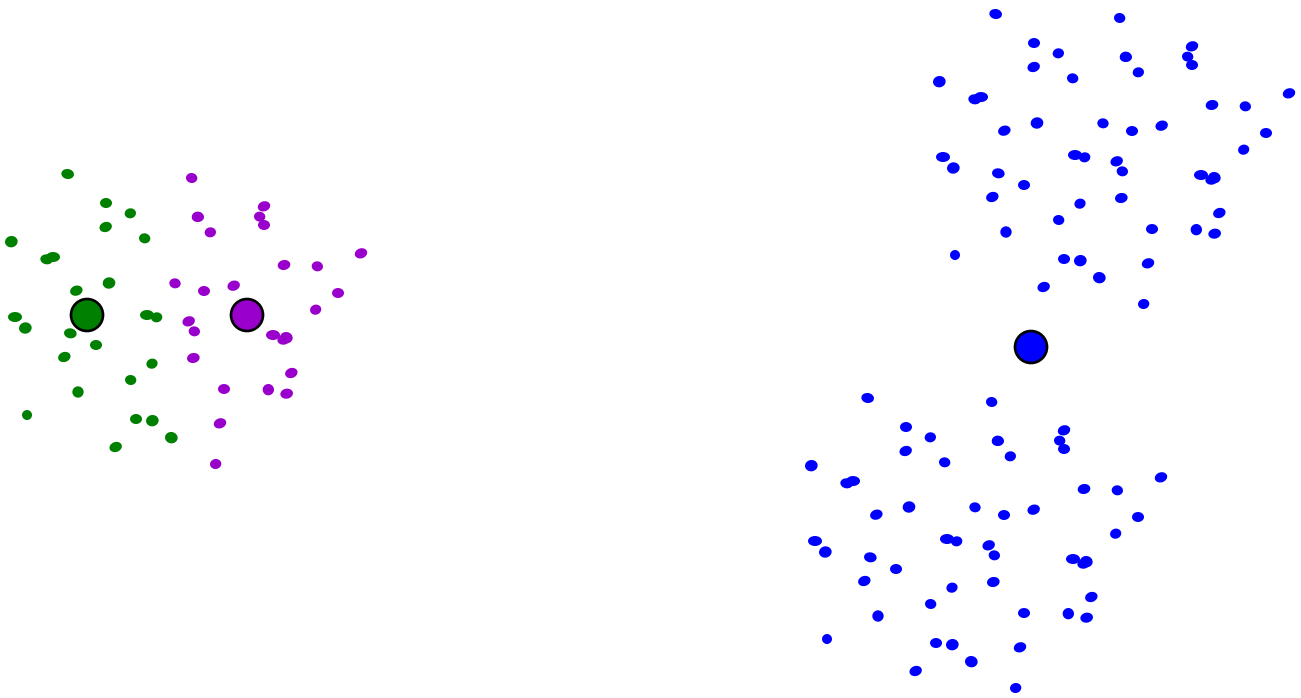
- K-means is sensitive to initialization
  - It does matter what you pick!
  - What can go wrong?
    - Various schemes to help alleviate this problem: initialization heuristics



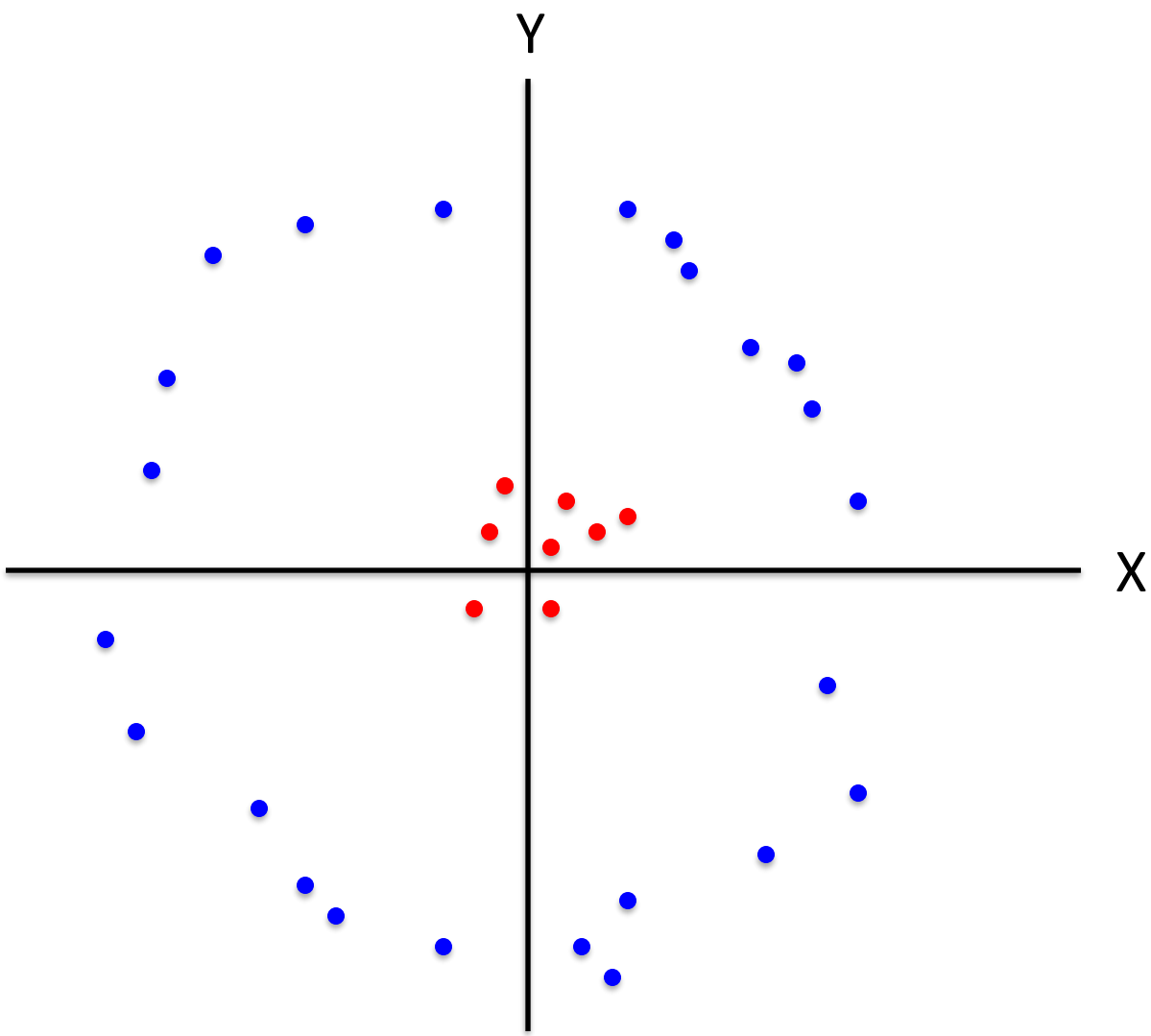
k-means Clustering

- Not clear how to figure out the "best" k in advance
- Want to choose k to pick out the interesting clusters, but not to overfit the data points
  - Large k doesn't necessarily pick out interesting clusters
  - Small k can result in large clusters than can be broken down further

Local Optima



Where k-means does not work



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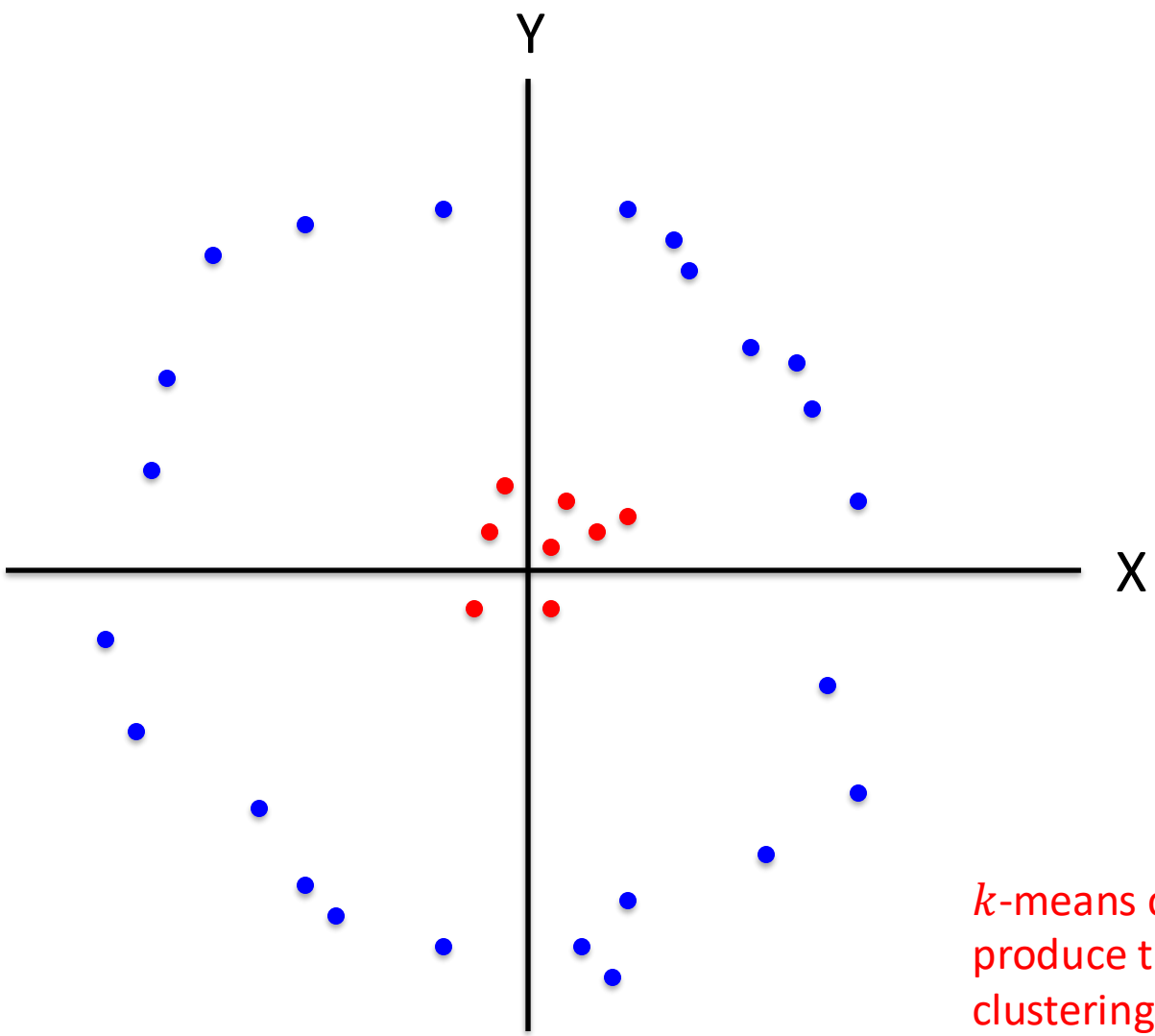
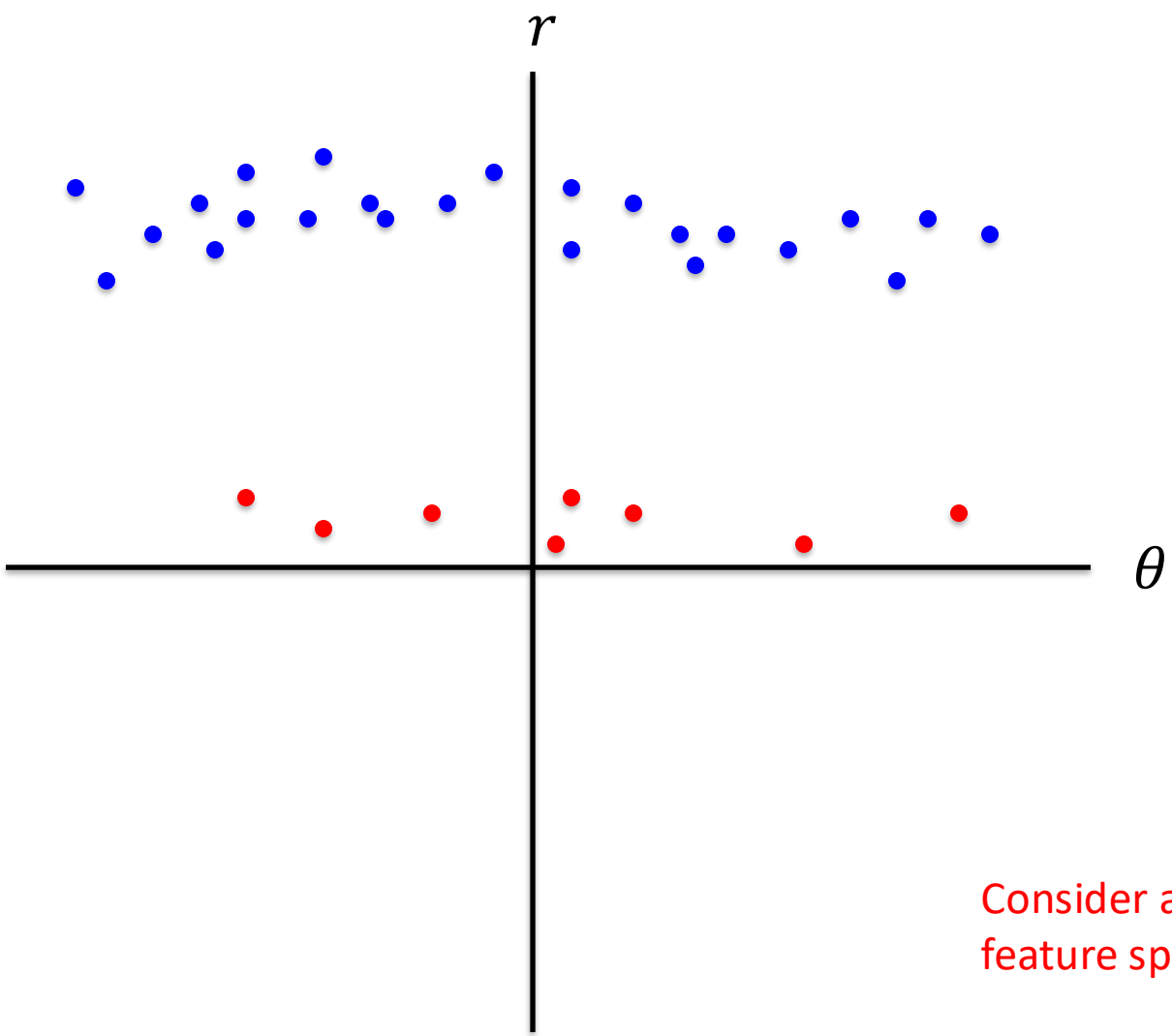


k-means cannot produce this kind of clustering!

How to make it work?



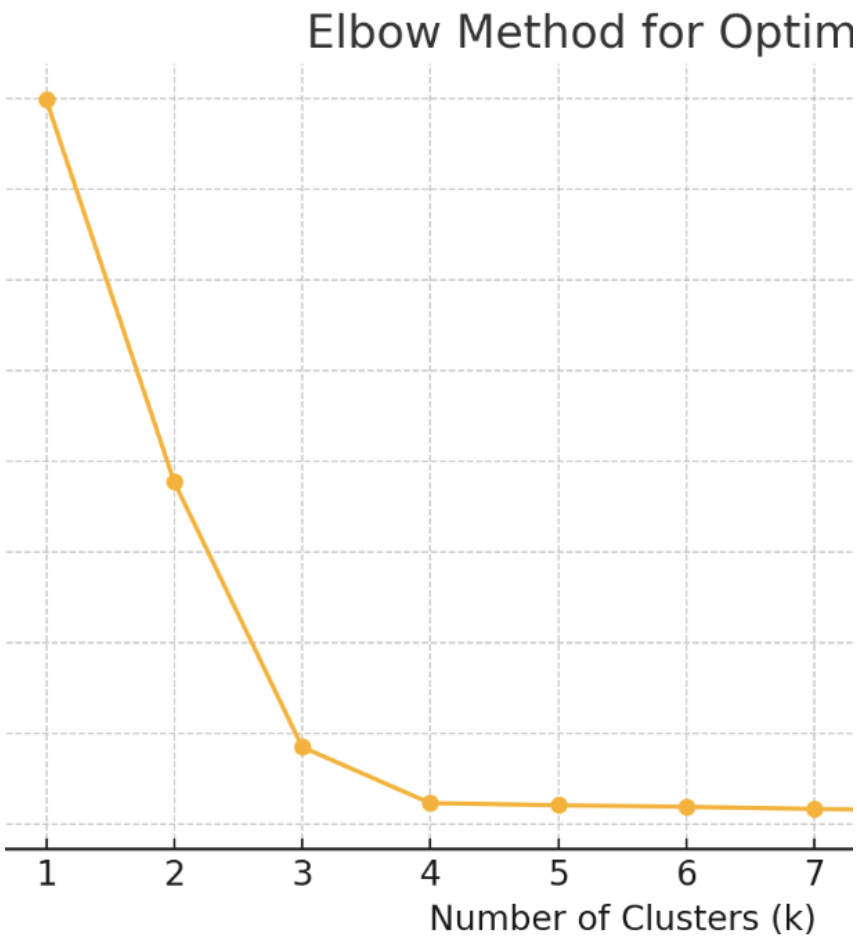
Consider alternative feature spaces...

k-Means Summary

- Guaranteed to converge
  - But not to a global optimum
- Choice of k and initialization can greatly affect the outcome
- Runtime: $O(kMn)$ per iteration
- Popular because it is fast, though there are other clustering methods that may be more suitable depending on your data

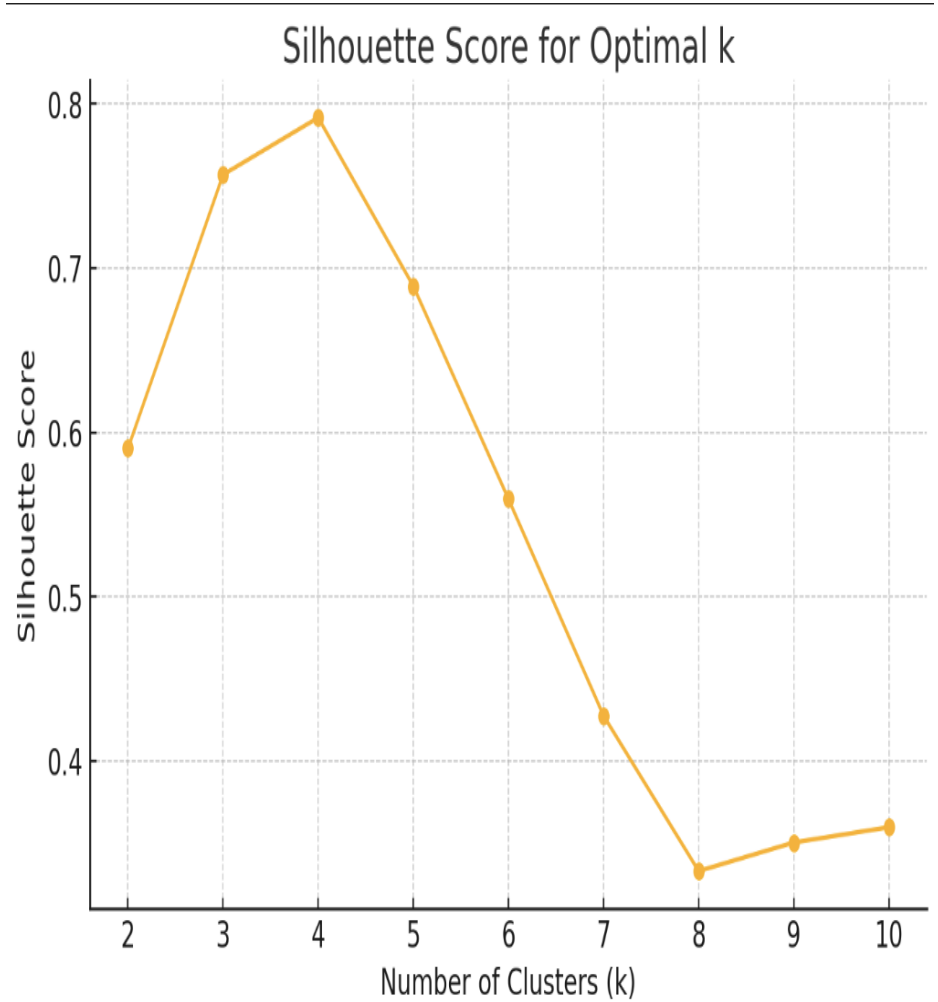
Selecting the Optimal K: Elbow Method

- The Elbow Method helps determine the optimal number of clusters by analyzing the inertia (within-cluster sum of squared distances).
- Inertia (SSE - Sum of Squared Errors): It measures how close data points are to their assigned cluster centers.
- Idea: If we plot inertia vs. the number of clusters (k), inertia decreases as k increases. However, there is a point where the decrease slows significantly—this is called the "elbow point", and it represents the optimal k.



Selecting the Optimal K: Silhouette Score

- The Silhouette Score measures how well-separated the clusters are. It evaluates: How close points in the same cluster are. How far apart different clusters are.
- $S(i) = b(i) - a(i) / \max(a(i), b(i))$
- Where:
  - a(i) is the average intra-cluster distance (**cohesion**).
  - b(i) is the average nearest-cluster distance (**separation**).



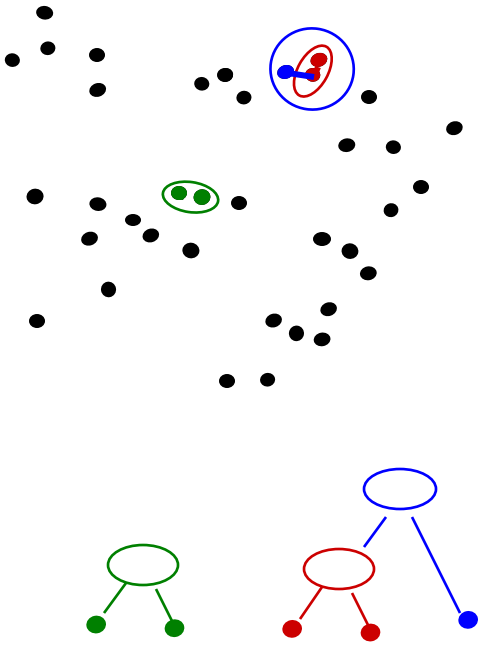
K-mediods Clustering and Extensions

- Very similar to k-means, except that the centroid is one of the examples from the cluster
- The update means step therefore involves finding the point within the cluster which has the minimum average distance to the rest

- Extensions of k-means: Other distance measures, e.g. the Bregman divergence

Hierarchical Clustering

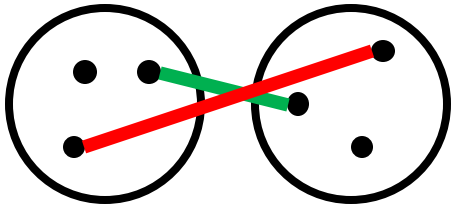
- Agglomerative clustering
  - Incrementally build larger clusters out of smaller clusters
- Algorithm:
  - Maintain a set of clusters
  - Initially, each instance in its own cluster
  - Repeat:
    - Pick the two closest clusters
    - Merge them into a new cluster
    - Stop when there is only one cluster left



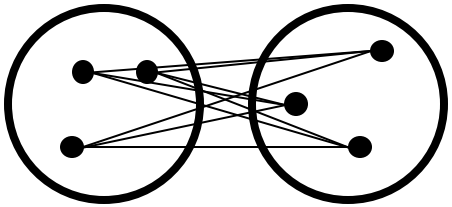
- Produces not one clustering, but a family of clusterings represented by a dendrogram

Agglomerative Clustering

- How should we define "closest" for clusters with multiple elements?



Closest / farthest pair

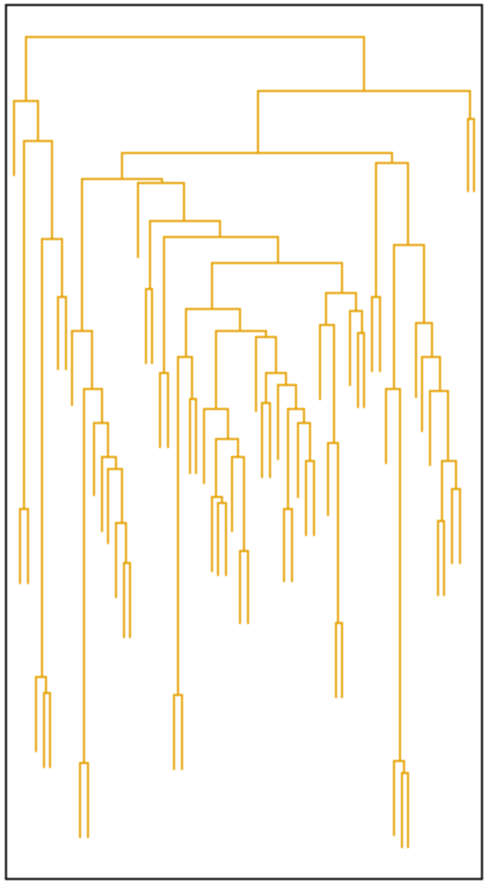


Average of all pairs

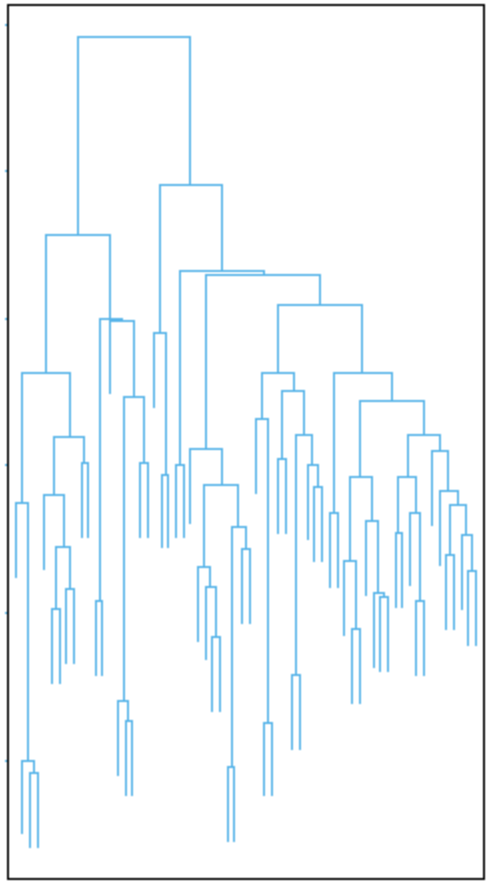
- Many more choices, each produces a different clustering...

Clustering Behavior

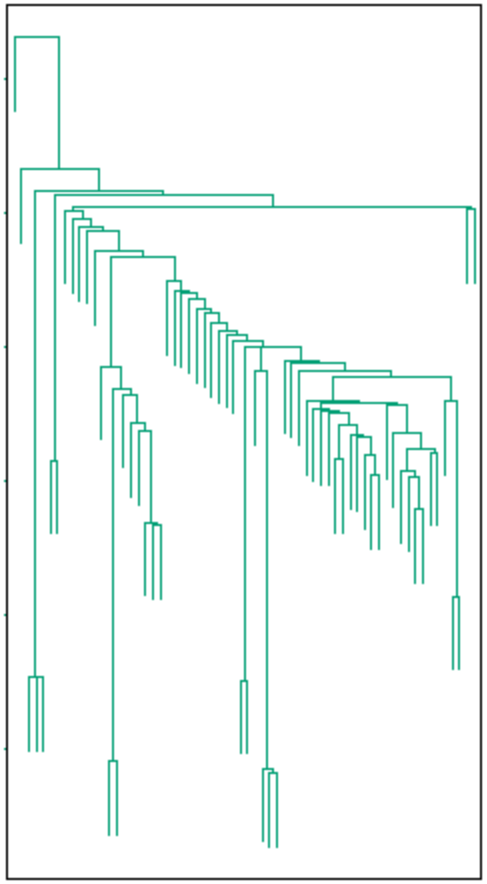
Average



Farthest



Nearest



Mouse tumor data from [Hastie]

Different kinds of Linkages

Linkage	Distance Considered	Cluster Shape	Best Used For	Limitations
Ward	Minimizes variance	Compact	Balanced clusters	Computationally expensive
Single	Closest point	Chain-like	Arbitrary shapes	Sensitive to noise
Complete	Farthest point	Compact & well-separated	Small, tight clusters	Large separation
Average	Mean distance	Moderate	General use	Can be affected by outliers

How to Choose the Right Linkage?

- Use **Ward** for compact, balanced clusters.
- Use **Single** if clusters have complex, non-convex shapes.
- Use **Complete** when you want well-separated, dense clusters.
- Use **Average** if you want a balance between compactness and flexibility.